

9 (**short**) A rigid circular disc of mass m and radius r is shown in Fig. 6. A point A is located on the edge of the disc at $\{x, y\} = r \{1, 1\}/\sqrt{2}$

(a) Find the moment of inertia I_{zz} at A. [5]

(b) A shaft parallel to the z axis is fixed to the disc at A. Find the angular acceleration of the disc when a torque Q is applied along the axis of this shaft. [5]

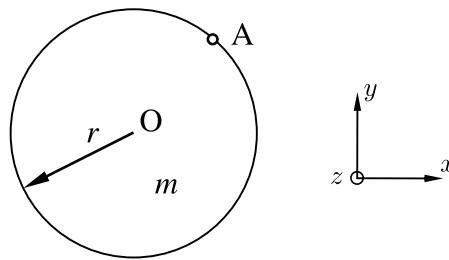


Fig. 6